



TRUEVISION: Vision Based Deepfake Detection System

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Abstract. As deep fake generation techniques evolve, detecting manipulated content across multiple modalities has become increasingly important to safeguard against their misuse. This paper presents a vision based deep fake detection system that leverages the power of pre-trained transformer models to analyze two different data modalities, facial images. For facial images, an image transformer captures pixel-level, structural and spatial inconsistencies. We discuss the integration of these transformers within a unified system, the synergistic effect of vision-based analysis, and the improvement in performance compared to CNN based approaches providing insights into potential directions for further enhancing the detection of sophisticated deep fakes

Keywords: Deep Fake Detection, Vision Based Analysis, Real-time Detection, Continuous Learning, BI, GAN, Grad-CAM, MCL, SOTA, NFA, Cosine Annealing, AdamW, Mediapipe.

1 Introduction

Deepfake technology has rapidly evolved, enabling the creation of highly realistic fake content that can be misused for misinformation and impersonation. The growing variety of generation techniques, such as GANs and diffusion models, challenges traditional detection methods. This research addresses these challenges by detecting manipulation-induced inconsistencies and artifacts in image data to improve robustness against evolving Deepfakes.

In this paper, we focus on transformer-based pre-trained architectures, which excel at handling complex visual inputs. The study also explores key datasets, evaluation metrics, and benchmarking frameworks essential for assessing detection systems, while emphasizing the need for lightweight, scalable, and generalizable models. Thus, Sophisticated Vision Based Deep Fake Detection Systems represent a crucial role in preserving the authenticity of online content and maintaining trust in digital platforms.

Given the increasing accessibility and misuse of Deepfakes, this research responds to the urgent need to counter their societal impact and restore public trust in digital media. The proposed vision-based system provides a unified platform for training, fine-tuning, and evaluating transformer models on diverse datasets. It assesses performance using accuracy, precision, recall, confusion matrix, AUC, precision-recall, and

sensitivity-specificity metrics. Designed for continuous learning, the system adapts to unseen data and integrates various transformer models to effectively extract discriminative visual features.

2 Study Scope

In this proposed method, the system is designed that efficiently detects face shift modifications, manipulations, alterations, artificially generated images and provides Grad-CAM explainability which highlights all the regions of image that helped the model to conclude about whether the image is fake or real. Flask is used as a backend which has minimal downtime and has the capacity to handle large numbers of requests without crashing and provide a quick and reasonable response to the USERS who will interact with the system.

3 Related Work

Liu et al. [1] propose a Deepfake detection method called Multimodal Contrastive Learning (MCL), which captures both intra- and cross-modal forgery cues. The method employs a cross-modal contrastive learning strategy to align uni-modal and multi-modal representations, reducing modality gaps. A Noise-based Feature Augmentation (NFA) module is introduced to perturb audio-visual features and enhance generalization. Despite outperforming state-of-the-art methods, the approach requires substantial multimodal data and is sensitive to the quality of extracted features.

Lu et al. [2] propose a spatial-temporal model with a long-distance attention mechanism to capture both spatial and temporal forgery traces in Deepfakes. By framing detection as a fine-grained classification task, the model distinguishes subtle differences between real and fake faces. The long-distance attention enhances global context and local feature extraction, achieving state-of-the-art results across datasets. However, the method incurs high computational cost and relies on the choice of backbone network.

Ramadhani et al. [3] present a ViViT-based Deepfake detection system using facial landmark area images as input. It integrates Depthwise Separable Convolution and Convolutional Block Attention Module to extract features from these regions. Trained on an augmented Celeb-DF v2 dataset with 8,335 videos, the model achieves 87.18% accuracy and a 92.52% F1 score. Ablation studies confirm that landmark areas, DSC, and CBAM significantly boost performance.

Yu et al. [4] propose the Multiple Spatiotemporal Views Transformer (MSVT), which combines Local (LSV) and Global Spatiotemporal Views (GSV) to capture detailed spatiotemporal cues for Deepfake detection. LSV identifies dynamic inconsistencies, while a Global-Local Transformer (GLT) integrates multi-level features. MSVT outperforms state-of-the-art methods in extensive experiments. However, its high computational cost and sensitivity to temporal sampling strategies require further exploration.

Liao et al. [5] propose a Deepfake detection method based on facial muscle motions, proving effective on compressed videos from social networks. By analyzing muscle movements, the approach offers a unique perspective and achieves strong performance. However, its reliability depends on accurate facial landmark extraction, and the role of different feature fusion strategies warrants further study

Yang et al. [6] introduce AVoiD-DF, an audio-visual joint learning framework that detects Deepfakes by exploiting audio-visual inconsistencies. A Temporal-Spatial Encoder captures temporal-spatial features, while a Multi-Modal Joint-Decoder fuses and models inter-modal relationships. A Cross-Modal Classifier identifies manipulations based on intra- and inter-modal disharmony. The authors also present DefakeAVMiT, a new benchmark dataset for multimodal Deepfake detection.

C. Zhao et al. [7] propose ISTVT, an Interpretable Spatial-Temporal Video Transformer for Deepfake detection that visualizes discriminative spatial and temporal regions. The model decomposes self-attention into spatial and temporal components to enhance interpretability. It achieves effective detection performance but faces potential limitations from high computational cost and the need to explore different self-attention mechanisms.

Z. Yin et al. [8] propose ID3, an Invariant Domain-oriented Deepfake Detection method that enhances generalization using invariant risk minimization. It employs Domain Refinement Data Augmentation and Domain Calibration Batch Normalization to suppress domain-specific features and align inputs into invariant representations. Experiments show improved generalization, but the method may require large, diverse datasets and further evaluation of augmentation strategies.

D. Liu et al. [9] proposes a hierarchical Deepfake detection method that learns robust patch-based hybrid domain representations for multi-modality scenarios. It extracts discriminative clues from both image and frequency domains, with a hierarchical classifier integrating these features for improved forgery detection. While effective, the method relies on accurate face landmark detection and requires further evaluation of its generalization across different forgery techniques.

Sohail Ahmed Khan et al. [10] analyze Deepfake detection generalization across model architectures, pre-training strategies, and datasets. Through intra- and inter-dataset evaluations on eight CNNs and two self-supervised transformers (DINO, CLIP), they find transformers outperform CNNs, with FaceForensics++ and DFDC providing superior generalization compared to FakeAVCeleb and CelebDF-V2. Image augmentation effects are also evaluated.

4 Methodology

4.1 Image Modality (Facial Deep Fake Detection)

The image modality processing pipeline consists of the following steps: Face Detection. We use MediaPipe as a face detection tool to verify if the uploaded image contains a clear and visible face such that the Transformer model can distinguish between fake and real images effectively.

MediaPipe serves as a filtering mechanism to determine if the image meets basic requirements for processing. The presence of face is detected and it is ensured that face size fulfills the criteria mentioned. ($>64 \times 64$). Then the image is checked upon various criteria such as exposure, blurriness etc.

Pretrained Transformers. We utilize two transformers specifically:

- Vision Transformer (ViT base patch16 224)
- Swin Transformer (Microsoft/swin-base-patch4-window7-224)

These transformers are effective for recognizing fine manipulations in facial features and textures.

Training and Fine-tuning. We combined multiple datasets including:

1. One Million Fake Faces <https://www.kaggle.com/datasets/tunguz/1-million-fake-faces>
2. Real(deepfake) dataset: <https://www.kaggle.com/datasets/victorkingau/deepfake>
3. Deepfake-dataset: <https://www.kaggle.com/datasets/aryansingh16/deepfakedataset>
4. Dataset for deepfake: <https://www.kaggle.com/datasets/tusharpadhy/real-dataset-for-deepfake-test>
5. Deepfake and real images: <https://www.kaggle.com/datasets/manjilkarki/deepfake-and-real-images>
6. Deep fake face detection: <https://www.kaggle.com/datasets/vasubhut/deep-fake-face-detection>

And final dataset consisted of 375,000 images that were split into train, validation and test in the ratio of 65,15,20. The model is fine-tuned on facial images for binary classification (real vs fake) by retraining the final classification layer in stage 1 for 15 epochs and the task-specific layers in stage 2 for 25 epochs. Data augmentation techniques (Flip, Rotation, Color Jitter, RandomResize-Crop, Normalize, toTensor, Affine, Perspective, SharpnessAdjust) are applied to ensure models see a variety of training samples. Weight decay (L2 ridge regularization) is used along with label smoothing to prevent overfitting and ensure good generalization on unseen data. We use Cosine Annealing Warm Restarts and Reduce LR on Plateau schedulers which adjust learning rate dynamically per epoch to ensure stable and effective convergence. AdamW optimizer is used, and the loss function uses class weights to handle minor class imbalance (55.28:44.72, fake: real images). An early stopping mechanism is used on the validation set to monitor model improvement after every epoch.

Optimization & Evaluation: Based on optimal hyperparameters and learned weights, the model is evaluated on test data. Performance is monitored using AUC-ROC curve and score to measure model's ability to distinguish between Real and Fake classes, Precision-Recall Curve for imbalanced dataset scenarios, Sensitivity-Specificity curve. Confusion matrix providing complete insight into model performance including Accuracy, Precision, Recall, F1 score, Specificity, TNR, FPR, FNR, TP, TN, FP, FN. User uploaded images and results along with model metrics are stored in PostgreSQL database for effective analytics through BI.

5 System Architecture

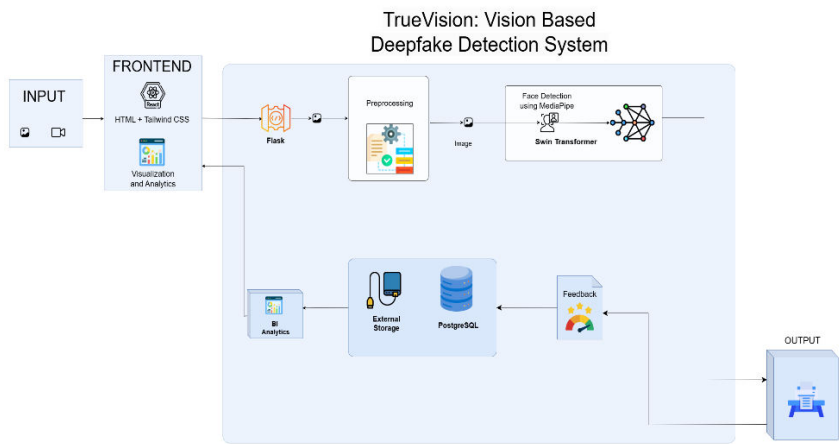


Fig. 1. System Architecture

The TrueVision system follows a modular architecture designed for detecting deepfake images using advanced deep learning techniques. The system begins with an input module that accepts images, processed through an HTML and CSS-based frontend for visualization and analytics, communicating with the backend via Flask. The preprocessing module extracts video frames and applies image normalization, resizing, and tensor conversion before feeding them into MediaPipe face detection. Detected faces are analyzed using ViT, MvT, Swin Transformer, deep learning models for deepfake classification. Results are stored in a PostgreSQL database with external storage for logging and future analysis. The architecture supports BI analytics for insights and reporting, delivering final deepfake detection output through a structured visualization interface.

6 Results and Analysis

6.1 Experimental Results

We conducted experiments in two stages: Stage-1 training (classifier only) and Stage-2 training (classifier + task-specific layer).

Table 1. Stage-1 Training Results (Classifier Only)

Model	Accur acy	Precis ion	Recall	F1 Score	AUC- ROC	PR- AUC
ResNet50	0.8108	0.7540	0.8675	0.8068	0.8974	0.8701
XceptionNet	0.7889	0.7744	0.7568	0.7655	0.8712	0.8376

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC	PR-AUC
EfficientB3	0.7973	0.7613	0.8084	0.7842	0.8804	0.8429
InceptionNetV3	0.7440	0.7281	0.6988	0.7132	0.8286	0.7908
DenseNet121	0.7858	0.7413	0.8137	0.7758	0.8682	0.8266
ViT	0.7883	0.7385	0.8286	0.7810	0.8718	0.8278
Swin	0.8641	0.8224	0.8947	0.8570	0.9375	0.9152

Table 2. Stage-2 Training Results (Classifier + Task-Specific Layer)

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC	PR-AUC
ResNet50	0.8196	0.7952	0.8135	0.8042	0.9015	0.8749
XceptionNet	0.7863	0.7769	0.7445	0.7604	0.8707	0.8342
EfficientB3	0.8165	0.7801	0.8316	0.8050	0.8988	0.8688
ViT	0.8018	0.7495	0.8481	0.7958	0.8843	0.8425
Swin	0.8688	0.8272	0.8999	0.8620	0.9410	0.9189

From both tables, we can conclude that the Swin Transformer consistently outperforms other models across all metrics. Even after fine-tuning the classifier and task-specific layers, the performance improvements are marginal, indicating that the models may have reached a saturation point. The Swin Transformer achieves the highest accuracy of 86.88% with an AUC-ROC of 0.9410, demonstrating superior performance in deepfake detection.

6.2 Confusion Matrix

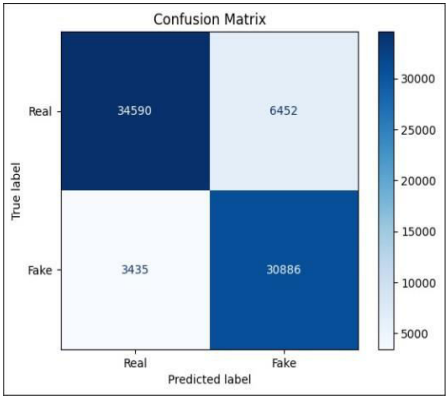


Fig. 2. Swin Confusion Matrix

6.3 System Implementation

Real Detection.

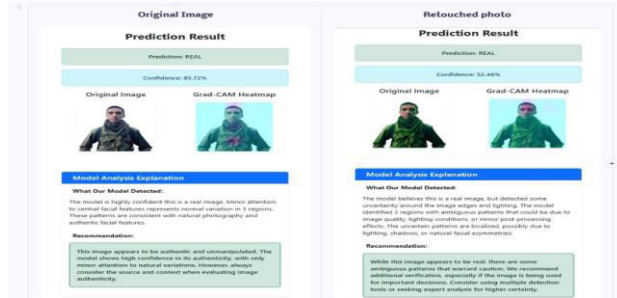


Fig. 3. Real Image

The image detected as Real in both scenarios. But the drastic change in the confidence score of the image on the right was the result of self-tampering of the image working on addition of noise, contrast, brightness, exposure to the image and other such factors. Highlights the model's ability to correctly detect subtle manipulations and changes with respect to the change in multiple factors leading to accurate predictions.

Fake Detection.

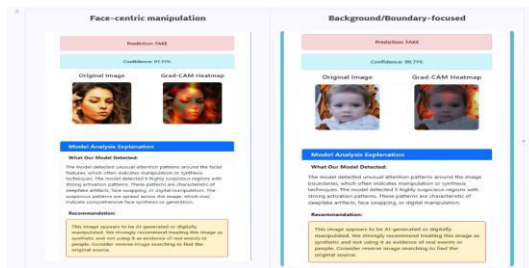


Fig. 4. Fake Image

The image detected as Fake in both scenarios. The left image highlights the scenario when models identify face-centric manipulation or tampered regions showing models ability to detect face-oriented features accurately. The right image highlights the scenario where models correctly identifies the background manipulations in the image with highlighting of regions models finds suspicious.

7 Conclusion and Future Work

By leveraging the power of transformer models and vision-based analysis, the TrueVision system achieves exceptional performance in detecting deep fakes across

visual media formats. The real-time detection capabilities and user-friendly interface further enhance its practicality and effectiveness.

Deep fake technology is constantly evolving, and new manipulation techniques emerge regularly. To stay ahead of constantly evolving manipulation techniques, self-improvisation and auto-updating of models are crucial. Future work will focus on inclusion of cross-model feature fusion enhancing the system's performance efficiency and comprehensive interpretation capabilities. Integration of audio signals or voice in videos will make the research more interesting. We also plan to use not only CNN and Transformer architectures but also experiment with GANs, AutoEncoders and Diffusion Networks along with different Video Transformers etc.

It is essential to use deep fake detection technology responsibly and ethically, ensuring that it is used to protect individuals and organizations from misinformation and manipulation while promoting trust and transparency in the digital world. The TrueVision system is a testament to the potential of technology to combat misinformation and protect the integrity of digital media. With continued research and development, we can create a safer and more trustworthy digital environment for all.

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